

Relations

Chapter 9

Relations and Their Properties

Section 9.1

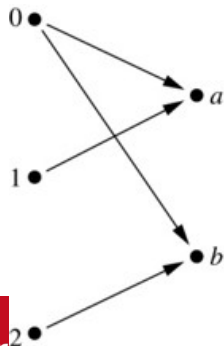
9.1.1 Introduction

Binary Relations(二元关系)

Definition: A *binary relation* R from a set A to a set B is a subset $R \subseteq A \times B$.

Example:

- Let $A = \{0,1,2\}$ and $B = \{a,b\}$
- $\{(0, a), (0, b), (1,a), (2, b)\}$ is a relation from A to B .
- We can represent relations from a set A to a set B graphically or using a table:



R	a	b
0	×	×
1	×	
2		×

Relations are more general than functions. A function is a relation where exactly one element of B is related to each element of A .

9.1.3 Binary Relations on a Set₁

Definition: A **binary relation R on a set A** is a subset of $A \times A$ or a relation from A to A .

Example:

- Suppose that $A = \{a, b, c\}$. Then $R = \{(a, a), (a, b), (a, c)\}$ is a relation on A .
- Let $A = \{1, 2, 3, 4\}$. The ordered pairs in the relation $R = \{(a, b) \mid a \text{ divides } b\}$
 $= \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$.

Binary Relations on a Set₃

Example: Consider these relations on the set of integers:

$$R_1 = \{(a, b) \mid a \leq b\},$$

$$R_4 = \{(a, b) \mid a = b\},$$

$$R_2 = \{(a, b) \mid a > b\},$$

$$R_5 = \{(a, b) \mid a = b + 1\},$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\},$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}.$$

Note that these relations are on an infinite set and each of these relations is an infinite set.

Which of these relations contain each of the pairs (1,1), (1, 2), (2, 1), (1, -1), and (2, 2)?

Solution: Checking the conditions that define each relation, we see that the pair (1,1) is in R_1, R_3, R_4 , and R_6 ; (1,2) is in R_1 and R_6 ; (2,1) is in R_2, R_5 , and R_6 ; (1, -1) is in R_2, R_3 , and R_6 ; (2,2) is in R_1, R_3 , and R_4 .

9.1.4 Properties of Relations

Reflexive Relations(自反关系)

Definition: A relation R on a set A is called *reflexive* if $(a, a) \in R$ for every element $a \in A$. Written symbolically, R is reflexive if and only if $\forall a (a \in A \rightarrow (a, a) \in R)$

Example: The following relations on the integers are reflexive:

$$R_1 = \{(a, b) \mid a \leq b\},$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\},$$

$$R_4 = \{(a, b) \mid a = b\}.$$

The following relations are not reflexive:

$$R_2 = \{(a, b) \mid a > b\} \quad (\text{note that } 3 \not> 3),$$

$$R_5 = \{(a, b) \mid a = b + 1\} \quad (\text{note that } 3 \neq 3 + 1),$$

$$R_6 = \{(a, b) \mid a + b \leq 3\} \quad (\text{note that } 4 + 4 > 3).$$

Symmetric Relations(对称关系)

Definition: A relation R on a set A is called *symmetric* if $(b, a) \in R$ whenever $(a, b) \in R$ for all $a, b \in A$. Written symbolically, R is symmetric if and only if $\forall x \forall y [(x, y) \in R \rightarrow (y, x) \in R]$

Example: The following relations on the integers are symmetric:

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\},$$

$$R_4 = \{(a, b) \mid a = b\},$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}.$$

The following are not symmetric:

$$R_1 = \{(a, b) \mid a \leq b\} \quad (\text{note that } 3 \leq 4, \text{ but } 4 \not\leq 3),$$

$$R_2 = \{(a, b) \mid a > b\} \quad (\text{note that } 4 > 3, \text{ but } 3 \not> 4),$$

$$R_5 = \{(a, b) \mid a = b + 1\} \quad (\text{note that } 4 = 3 + 1, \text{ but } 3 \neq 4 + 1).$$

Antisymmetric Relations(反对称关系)

Definition: A relation R on a set A such that for all $a, b \in A$ if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$ is called *antisymmetric*. Written symbolically, R is antisymmetric if and only if

$$\forall x \forall y [(x, y) \in R \wedge (y, x) \in R \rightarrow x = y]$$

Example: The following relations on the integers are antisymmetric:

$$R_1 = \{(a, b) \mid a \leq b\},$$

$$R_2 = \{(a, b) \mid a > b\},$$

$$R_4 = \{(a, b) \mid a = b\},$$

$$R_5 = \{(a, b) \mid a = b + 1\}.$$

The following relations are not antisymmetric:

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

(note that both $(1, -1)$ and $(-1, 1)$ belongs to R_3),

$$R_6 = \{(a, b) \mid a + b \leq 3\} \text{ (note that both } (1, 2) \text{ and } (2, 1) \text{ belongs to } R_6).$$

← For any integer, if a $a \leq b$ and $b \leq a$, then $a = b$.

R_2 is antisymmetric because it is impossible that $a > b$ and $b > a$.

R_4 is antisymmetric, because two elements are related with respect to R_4 if and only if they are equal.

R_5 is antisymmetric because it is impossible that $a = b + 1$ and $b = a + 1$.

Transitive Relations(传递关系)

Definition: A relation R on a set A is called **transitive** if whenever $(a,b) \in R$ and $(b,c) \in R$, then $(a,c) \in R$, for all $a,b,c \in A$. Written symbolically, R is transitive if and only if

$$\forall x \forall y \forall z \left[(x,y) \in R \wedge (y,z) \in R \rightarrow (x,z) \in R \right]$$

Example: The following relations on the integers are transitive:

$$R_1 = \{(a,b) \mid a \leq b\},$$

$$R_2 = \{(a,b) \mid a > b\},$$

$$R_3 = \{(a,b) \mid a = b \text{ or } a = -b\},$$

$$R_4 = \{(a,b) \mid a = b\}.$$

For every integer, $a \leq b$
and $b \leq c$, then $a \leq c$.

The following are not transitive:

$$R_5 = \{(a,b) \mid a = b + 1\} \quad (\text{note that } (2,1) \text{ and } (1,0) \text{ belong to } R_5, \text{ but } (2,0) \text{ does not.})$$

$$R_6 = \{(a,b) \mid a + b \leq 3\} \quad (\text{note that } (2,1) \text{ and } (1,2) \text{ belong to } R_6, \text{ but } (2,2) \text{ does not.})$$

$$\forall a (a \in A \rightarrow (a, a) \in R)$$

整数集合 $\mathbb{Z}$ 中，关系 $\leq$ 是自反的，而关系 $<$ 不是自反的

$$\forall x \forall y [(x, y) \in R \rightarrow (y, x) \in R]$$

在整数集合 $\mathbb{Z}$ 中，相等关系 $=$ 是对称的，而关系 $\leq$ 和 $<$ 都不是对称的。

$$\forall x \forall y [(x, y) \in R \wedge (y, x) \in R \rightarrow x = y]$$

该定义表明了，在表示反对称关系 R 的有序对集合中，若存在有序对 $\langle x, y \rangle$ 和 $\langle y, x \rangle$ ，则必定是 $x = y$ 。注意，若 R 中对于 $\langle x, y \rangle$ 处处不出现 $\langle y, x \rangle$ ，则仍满足反对称性(比方说 $<$)。在整数集合 $\mathbb{Z}$ 中， $=$ ， $\leq$ 和 $<$ 也都是反对称的。

$$\forall x \forall y \forall z [(x, y) \in R \wedge (y, z) \in R \rightarrow (x, z) \in R]$$

该定义表明了，在表示可传递关系 R 的有序对集合中，若有有序对 $\langle x, y \rangle$ 和 $\langle y, z \rangle$ ，则必有有序对 $\langle x, z \rangle$ 。

$\leq$ ， $<$ ， $=$ 都是传递的

反对称性是一个关于数学上二元关系的性质。大概地说，集合 X 上的二元关系 R 是反对称的，当且仅当 不存在 X 里的一对相异元素 a, b ，它们相互 R -关系于彼此

整数 上的整除关系 $|$ 不是反对称的（因为 $1|-1$ ， $-1|1$ ，但 $1 \neq -1$ ）。如果限制在自然数范围内则是反对称的。整数上的整除关系，不是对称的。

整数上的模 n 同余 是对称的，但不是反对称的。

这页只帮助理解 不要求会

注意，反对称关系不是对称关系 ($aRb \rightarrow bRa$) 的反义。有些关系既是对称的又是反对称的，比如“等于”。有些关系既不是对称的也不是反对称的，比如上面说的整数上整除例子。

非对称性 (not symmetric) ($aRb \wedge \sim bRa$) 才算是对称关系的反义。事实上，非对称性关系都符合反对称性 (antisymmetric)。 $\forall x \forall y [(x, y) \in R \wedge (y, x) \in R \rightarrow x = y]$

###

Asymmetric relation (非对称关系) For all a and b in X , if a is related to b , then b is not related to a . $\forall a, b \in X : aRb \rightarrow \neg(bRa)$.

An example of an asymmetric relation is the “less than” relation $<$ between real numbers: if $x < y$, then necessarily y is not less than x . The “less than or equal” relation $\leq$, on the other hand, is not asymmetric, because reversing e.g. $x \leq x$ produces $x \leq x$ and both are true. Asymmetry is not the same thing as “not symmetric”: the less-than-or-equal relation is an example of a relation that is neither symmetric nor asymmetric.

考虑整数集合中	Reflexive Relations(自反关系)	Symmetric Relations(对称关系)	Antisymmetric Relations(反对称关系)	Transitive Relations(传递关系)
	$\forall a (a \in A \rightarrow (a, a) \in R)$	$\forall x \forall y [(x, y) \in R \rightarrow (y, x) \in R]$	$\forall x \forall y \forall z [(x, y) \in R \wedge (y, z) \in R \rightarrow (x, z) \in R]$	
$\leq$	Yes	Not	Yes	Yes
$<$	Not	Not	Yes	Yes
$=$	Yes	Yes	Yes	Yes
整數上Divisible relation 整除关系	Yes	Not	Not	Yes
Congruence modulo n 模n同余	Yes	Yes	Not	Yes

9.1.5 Combining Relations(合并关系)

Given two relations R_1 and R_2 , we can combine them using basic set operations to form new relations such as $R_1 \cup R_2$, $R_1 \cap R_2$, $R_1 - R_2$, and $R_2 - R_1$.

Example: Let $A = \{1,2,3\}$ and $B = \{1,2,3,4\}$. The relations $R_1 = \{(1,1),(2,2),(3,3)\}$ and $R_2 = \{(1,1),(1,2),(1,3),(1,4)\}$ can be combined using basic set operations to form new relations:

$$R_1 \cup R_2 = \{(1,1),(1,2),(1,3),(1,4),(2,2),(3,3)\}$$

$$R_1 \cap R_2 = \{(1,1)\} \quad R_1 - R_2 = \{(2,2),(3,3)\}$$

$$R_2 - R_1 = \{(1,2),(1,3),(1,4)\}$$

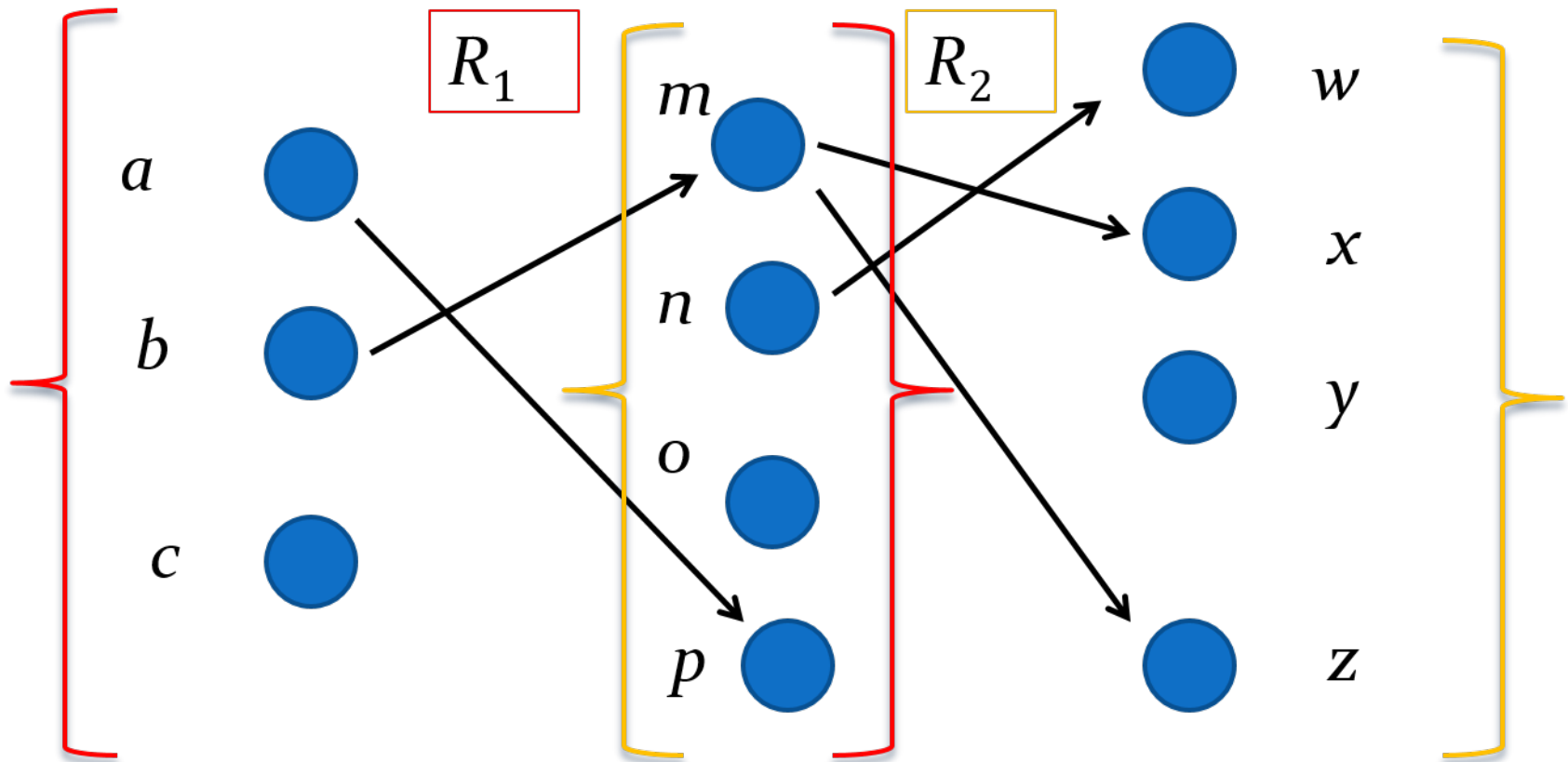
Composition(合成关系)

Definition: Suppose

- R is a relation from a set A to a set B .
- S is a relation from B to a set C .

The *composite* of R and S is the relation consisting of ordered pairs (a, c) , where $a \in A$, $c \in C$, and for which there exists an element $b \in B$ such that $(a, b) \in R$ and $(b, c) \in S$. We denote the composite of R and S by $S \circ R$.

Representing the Composition of Relations



$$R_2 \circ R_1 = \{(b, x), (b, z)\}$$

Powers of a Relation

Definition: Let R be a binary relation on A . Then the powers R^n of the relation R can be defined recursively by:

- Basis Step: $R^1 = R$
- Recursive Step: $R^{n+1} = R^n \circ R$

The definition shows that $R^2 = R \circ R$, $R^3 = R^2 \circ R = (R \circ R) \circ R$, and so on.

Example: Let $R = \{(1,1), (2,1), (3,2), (4,3)\}$. Find the powers R^n , $n = 2, 3, 4, \dots$

Solutions: Because $R^2 = R \circ R$, we find that $R^2 = \{(1,1), (2,1), (3,1), (4,2)\}$. Furthermore, because $R^3 = R^2 \circ R$, $R^3 = \{(1,1), (2,1), (3,1), (4,1)\}$. Additional computation shows that R^4 is the same as R^3 , so $R^4 = \{(1,1), (2,1), (3,1), (4,1)\}$. It also follows that $R^n = R^3$ for $n = 5, 6,$

Powers of a Relation

The powers of a **transitive relation** are subsets of the relation.
This is established by the following theorem:

Theorem 1: The relation R on a set A is transitive iff

$$R^n \subseteq R \text{ for } n = 1, 2, 3, \dots$$

定理 设 R 为 A 上的关系, 则

$$R \text{ 在 } A \text{ 上传递} \Leftrightarrow R \circ R \subseteq R$$

证明 必要性: 任取 $\langle x, y \rangle$ 有

$$\begin{aligned} &\langle x, y \rangle \in R \circ R \\ \Rightarrow &\exists t (\langle x, t \rangle \in R \wedge \langle t, y \rangle \in R) \\ \Rightarrow &\langle x, y \rangle \in R \text{ (因为 } R \text{ 在 } A \text{ 上是传递的)} \end{aligned}$$

所以 $R \circ R \subseteq R$ 。

充分性: 任取 $\langle x, y \rangle, \langle y, z \rangle \in R$, 则

$$\begin{aligned} &\langle x, y \rangle \in R \wedge \langle y, z \rangle \in R \\ \Rightarrow &\langle x, z \rangle \in R \circ R \\ \Rightarrow &\langle x, z \rangle \in R \text{ (因为 } R \circ R \subseteq R) \end{aligned}$$

所以 R 在 A 上是传递的。

Section Summary

Relations *binary relation* $R \subseteq A \times B$.

Properties of Relations

- Reflexive Relations $\forall a (a \in A \rightarrow (a, a) \in R)$
- Symmetric and Antisymmetric Relations
 $\forall x \forall y [(x, y) \in R \rightarrow (y, x) \in R]$
 $\forall x \forall y [(x, y) \in R \wedge (y, x) \in R \rightarrow x = y]$
- Transitive Relations
 $\forall x \forall y \forall z [(x, y) \in R \wedge (y, z) \in R \rightarrow (x, z) \in R]$

Combining Relations $R_1 \cup R_2, R_1 \cap R_2, R_1 - R_2$, and $R_2 - R_1$.

Composition $S \circ R$

exercise

1. List the ordered pairs in the relation R from $A = \{0, 1, 2, 3, 4\}$ to $B = \{0, 1, 2, 3\}$, where $(a, b) \in R$ if and only if

a) $a = b$.

b) $a + b = 4$.

c) $a > b$.

d) $a \mid b$.

32. Let R be the relation $\{(1, 2), (1, 3), (2, 3), (2, 4), (3, 1)\}$, and let S be the relation $\{(2, 1), (3, 1), (3, 2), (4, 2)\}$. Find $S \circ R$.

30. Let $R_1 = \{(1, 2), (2, 3), (3, 4)\}$ and $R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), (3, 4)\}$ be relations from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$. Find

b) $R_1 \cap R_2$.

d) $R_2 - R_1$.

7. Determine whether the relation R on the set of all integers is reflexive, symmetric, antisymmetric, and/or transitive, where $(x, y) \in R$ if and only if

a) $x \neq y$.

b) $xy \geq 1$.

c) $x = y + 1$ or $x = y - 1$.

d) $x \equiv y \pmod{7}$.

e) x is a multiple of y .

Representing Relations

Section 9.3

9.3.2 Representing Relations Using Matrices

A relation between finite sets can be represented using a **zero-one matrix**.

Suppose R is a relation from $A = \{a_1, a_2, \dots, a_m\}$ to $B = \{b_1, b_2, \dots, b_n\}$.

- The elements of the two sets can be listed in any particular arbitrary order. When $A = B$, we use the same ordering.

The relation R is represented by the matrix $M_R = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R, \\ 0 & \text{if } (a_i, b_j) \notin R. \end{cases}$$

The matrix representing R has a **1** as its (i,j) entry **when a_i is related to b_j** and a **0** if **a_i is not related to b_j** .

Examples of Representing Relations Using Matrices₁

Example 1: Suppose that $A = \{1,2,3\}$ and $B = \{1,2\}$. Let R be the relation from A to B containing (a,b) if $a \in A$, $b \in B$, and $a > b$. What is the matrix representing R (assuming the ordering of elements is the same as the increasing numerical order)?

Solution: Because $R = \{(2,1), (3,1), (3,2)\}$, the matrix is $M_R = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$

Example 2: Let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4, b_5\}$. Which ordered pairs are in the relation R represented by the matrix

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix} ?$$

- Solution:** Because R consists of those ordered pairs (a_i, b_j) with $m_{ij} = 1$, it follows that:

$$R = \{(a_1, b_2), (a_2, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_1), (a_3, b_3), (a_3, b_5)\}.$$

Matrices of Relations on Sets

The matrix of a relation **on a set**, which is a square matrix, can be used to determine whether the relation has certain properties.

If R is a **reflexive** relation, all the elements on the **main diagonal of M_R** are equal to 1. Off diagonal elements can be 0 or 1.

$$\forall a (a \in A \rightarrow (a, a) \in R)$$

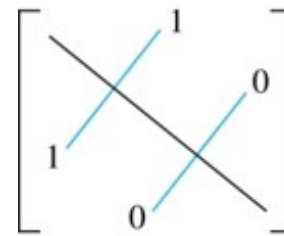
R is a **symmetric** relation, if and only if $m_{ij} = 1$ whenever $m_{ji} = 1$. R is symmetric if and only if **M_R is a symmetric matrix**. Diagonal elements can be 0 or 1.

$$\forall x \forall y [(x, y) \in R \rightarrow (y, x) \in R]$$

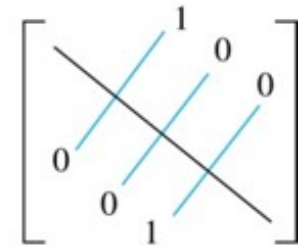
R is an **antisymmetric** relation, if and only if $m_{ij} = 0$ or $m_{ji} = 0$ when $i \neq j$.

Diagonal elements can be 0 or 1.

$$\forall x \forall y [(x, y) \in R \wedge (y, x) \in R \rightarrow x = y]$$



(a) Symmetric



(b) Antisymmetric

Example of a Relation on a Set

Example 3: Suppose that the relation R on a set is represented by the matrix

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Is R reflexive, symmetric, and/or antisymmetric?

Solution: Because all the diagonal elements are equal to 1, R is reflexive. Because M_R is symmetric, R is symmetric and not antisymmetric because both $m_{1,2}$ and $m_{2,1}$ are 1.

The Boolean operations join and meet (discussed in Section 2.6) can be used to find the matrices representing the union and the intersection of two relations. Suppose that R_1 and R_2 are relations on a set A represented by the matrices M_{R_1} and M_{R_2} , respectively. The matrices representing the union and intersection of these relations are

$$\mathbf{M}_{R_1 \cup R_2} = \mathbf{M}_{R_1} \vee \mathbf{M}_{R_2} \quad \text{and} \quad \mathbf{M}_{R_1 \cap R_2} = \mathbf{M}_{R_1} \wedge \mathbf{M}_{R_2}.$$

Boolean operations $\vee$ Join $\wedge$ meet

Example

Suppose that the relations R_1 and R_2 on a set A are represented by the matrices

$$\mathbf{M}_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

What are the matrices representing $R_1 \cup R_2$ and $R_1 \cap R_2$?

Solution: The matrices of these relations are

$$\mathbf{M}_{R_1 \cup R_2} = \mathbf{M}_{R_1} \vee \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix},$$

$$\mathbf{M}_{R_1 \cap R_2} = \mathbf{M}_{R_1} \wedge \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

We now turn our attention to determining the matrix for the **composite of relations**. This matrix can be found using **the Boolean product of the matrices** (discussed in Section 2.6) for these relations. In particular, suppose that R is a relation from A to B and S is a relation from B to C . Suppose that A , B , and C have m , n , and p elements, respectively. Let the zero–one matrices for $S \circ R$, R , and S be $M_{S \circ R} = [t_{ij}]$, $M_R = [r_{ij}]$, and $M_S = [s_{ij}]$, respectively (these matrices have sizes $m \times p$, $m \times n$, and $n \times p$, respectively).

$$\mathbf{M}_{S \circ R} = \mathbf{M}_R \odot \mathbf{M}_S.$$

$$\mathbf{M}_{R^2} = \mathbf{M}_R^{[2]}$$

Example

Find the matrix representing the relations $S \circ R$, where the matrices representing R and S are

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Solution: The matrix for $S \circ R$ is

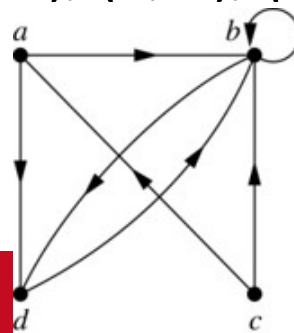
$$\mathbf{M}_{S \circ R} = \mathbf{M}_R \odot \mathbf{M}_S = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

9.3.3 Representing Relations Using Digraphs

Definition: A *directed graph* (有向图), or *digraph*, consists of a set V of *vertices* (顶点) (or *nodes* 节点) together with a set E of *ordered pairs of elements of V* called *edges* (边) (or *arcs* 弧线). The vertex (顶点) a is called the *initial vertex* of the edge (a, b) , and the vertex b is called the *terminal vertex* of this edge.

- An edge of the form (a, a) is represented using an arc from the vertex a back to itself. Such an edge is called a *loop* (环).

Example 7: A drawing of the directed graph with vertices a, b, c , and d , and edges (a, b) , (a, d) , (b, b) , (b, d) , (c, a) , (c, b) , and (d, b) is shown here.



Determining which Properties a Relation has from its Digraph

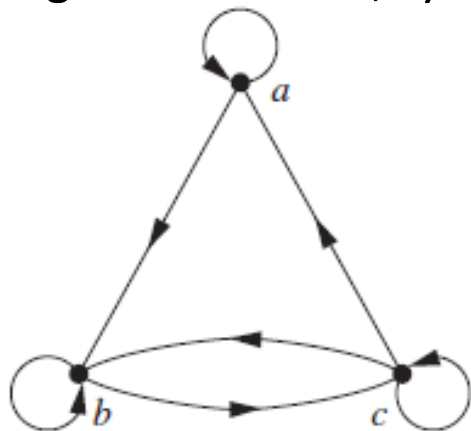
Reflexivity: A loop must be present at all vertices in the graph.

Symmetry: If (x,y) is an edge, then so is (y,x) .

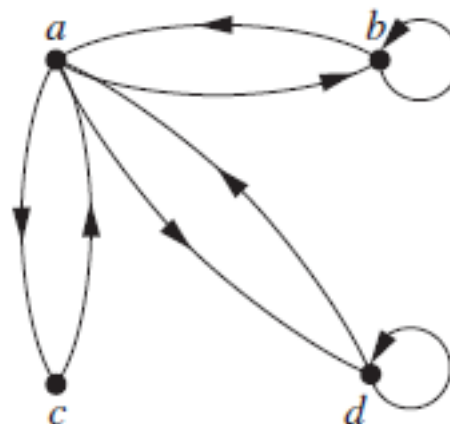
Antisymmetry: If (x,y) with $x \neq y$ is an edge, then (y,x) is not an edge.

Transitivity: If (x,y) and (y,z) are edges, then so is (x,z) .

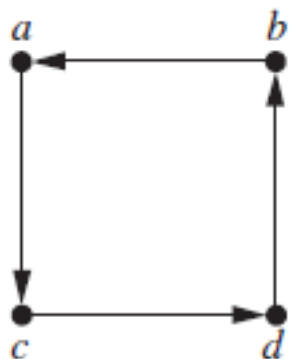
Determine whether the relations for the directed graphs shown in following figures reflexive, symmetric, antisymmetric, and/or transitive?



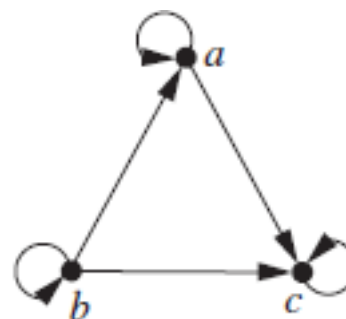
Reflexive, not symmetric, not antisymmetric,
Not transitive.



Symmetric, not reflexive, not
antisymmetric, not transitive



antisymmetric, not reflexive,
not symmetric, not transitive,
Because learning changes everything.™



Reflexive, antisymmetric, transitive,
not symmetric

因为关系的性质如传递性不仅反映在它的集合表达式上，也明显地反映在它的关系矩阵和它的关系图上。见表 1

表 1 传递性的各种表达

性质 表示	传递性
集合表达式	$R \circ R \subseteq R$
关系矩阵	$M^2 \subseteq M$
关系图	如果顶点 x_i 到 x_j 有边， x_j 到 x_k 有边，则从 x_i 到 x_k 有边

3.2 方法二：复合矩阵法

设 M 是 R 的关系矩阵， $M[i, j]$ 表示第 i 行第 j 列的元素。根据上面定理即 R 具有传递性 $\Leftrightarrow M^2 \subseteq M$ ，只要考查关系矩阵 M 中的零元素，当 $a_{ij} = 0$ 时，看 M^2 中对应位置的 b_{ij} 是否为零，若 $b_{ij} = 0$ ，这说明 R 是传递的，若 $b_{ij} = 1$ ，则 R 不是传

3.2 方法二：复合矩阵法

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如例 1 题得

$$M = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$
$$M^2 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

由复合矩阵判断法得此关系是可传递的。
此种方法尤其适用于零元素个数较少时的判断。

Section Summary

1, A relation between finite sets can be represented using a **zero-one matrix**.

$$\begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 & \\ & & & & & 1 \end{bmatrix}$$

reflexive

$$\begin{bmatrix} & & 1 & \\ & & & 0 \\ 1 & & & \\ & 0 & & \end{bmatrix}$$

(a) **Symmetric**

$$\begin{bmatrix} & & 1 & 0 \\ & & & 0 \\ 0 & & & \\ 0 & 1 & & \end{bmatrix}$$

(b) **Antisymmetric**

2, The **Boolean operations join and meet** (discussed in Section 2.6) can be used to find the **matrices representing the union and the intersection of two relations**.

$$\mathbf{M}_{R_1 \cup R_2} = \mathbf{M}_{R_1} \vee \mathbf{M}_{R_2} \quad \text{and} \quad \mathbf{M}_{R_1 \cap R_2} = \mathbf{M}_{R_1} \wedge \mathbf{M}_{R_2}.$$

3, We now turn our attention to determining the matrix for the **composite of relations**. This matrix can be found using **the Boolean product of the matrices** for these relations.

$$\mathbf{M}_{S \circ R} = \mathbf{M}_R \odot \mathbf{M}_S.$$

Determining which Properties a Relation has from its Digraph

Reflexivity: A loop must be present at all vertices in the graph.

Symmetry: If (x,y) is an edge, then so is (y,x) .

Antisymmetry: If (x,y) with $x \neq y$ is an edge, then (y,x) is not an edge.

Transitivity: If (x,y) and (y,z) are edges, then so is (x,z) .

Exercise

1. Represent each of these relations on $\{1, 2, 3\}$ with a matrix (with the elements of this set listed in increasing order).

a) $\{(1, 1), (1, 2), (1, 3)\}$

b) $\{(1, 2), (2, 1), (2, 2), (3, 3)\}$

3. List the ordered pairs in the relations on $\{1, 2, 3\}$ corresponding to these matrices (where the rows and columns correspond to the integers listed in increasing order).

a)
$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

b)
$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

7. Determine whether the relations represented by the matrices in Exercise 3 are reflexive, irreflexive, symmetric, antisymmetric, and/or transitive.

4. List the ordered pairs in the relations on $\{1, 2, 3, 4\}$ corresponding to these matrices (where the rows and columns correspond to the integers listed in increasing order).

a)
$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

15. Let R be the relation represented by the matrix

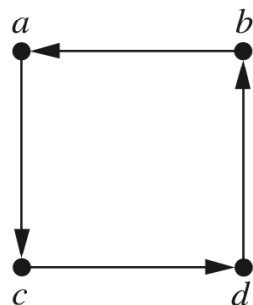
$$\mathbf{M}_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Find the matrices that represent

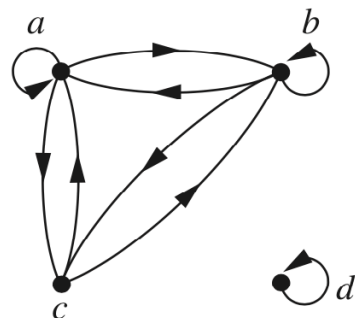
a) R^2 .

In Exercises 23–28 list the ordered pairs in the relations represented by the directed graphs.

25.



27.



Determine whether the relations represented by the directed graph shown in Exercise 25 27 are reflexive, symmetric, antisymmetric, and/or transitive

Homework

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Determine whether the relations represented by the directed graph shown in Exercise 23 26 are reflexive, symmetric, antisymmetric, and/or transitive.